

Agents

- A - Alice
- B - Browser with Session storage, SubtleCrypto API, and libsodium library
- M - Webauthn authenticator (passkey manager)
- S - Quick Crypt server

Browser and Libsodium Functions

- D_H - BLAKE2b-512 KDF from libsodium
- D_P - PBKDF2-HMAC-SHA512 key derivation FIPS-180-4 from SubtleCrypto
- D_S - HMAC-SHA-512 from SubtleCrypto
- E_a - Symmetric cipher using AEAD algorithm a . One of:
 - 1. AES-256 in Galois/Counter mode from SubtleCrypto
 - 2. XChaCha20 with Poly1305 MAC from libsodium
 - 3. AEGIS 256 from libsodium
- G - HMAC-SHA-512 key generator from SubtleCrypto
- H - BLAKE2b keyed hash (MAC) generator from libsodium
- R - Cryptographic pseudo-random generator from libsodium
- V - Constant-time hash (MAC) validator from libsodium

Cipher Variables

a - Symmetric AEAD cipher and mode: [1, 2, 3]
ad_F - Additional data included in ciphertext
ad_X - *ad_F* with extended context
b - Valid or invalid MAC tag
cd - Cipher data
cd₀ - Cipher data block 0
cd_N - Cipher data block N
err - Error message and exit
f - Block flags
h - Password hint text
h_E - Encrypted hint
h_El - Encrypted hint length
i - PBKDF2-HMAC-SHA512 iteration count: min 420,000 max 4,294,000,000
k_H - 256 bit ephemeral hint cipher key
k_{M0} - 256 bit ephemeral message block0 cipher key
k_{MN} - 256 bit ephemeral message blockN cipher key
k_S - 256 bit ephemeral MAC key
k_C - 256 bit ephemeral commit key
kp_B - Key purpose text: "Blck_Key"
kp_C - Key purpose text: "Cmit_Key"
kp_H - Key purpose text: "Hint_Key"
kp_S - Key purpose text: "Sign_Key"
l - Payload length
le - Loop end (1-16)
lp - Loop count (1-16)
m - Clear text
m₀ - Clear text block 0
m_N - Clear text block N
m_E - Block of encrypted message
n_{IV} - Pseudo random initialization vector
n_{IV}l - *n_{IV}* bit length: [96, 192, 256]
n_S - 128 bit pseudo random salt
N - Block number
p - Password text
r - 384 bits of pseudo random data
t - 256 bit MAC tag
t_L - Last 256 bit MAC tag
u_c - 256 bit user credential
v - Cipher data version: 7

Message Encryption by A

$A \xleftrightarrow{\text{webauthn}} B, M \xleftrightarrow{\text{webauthn}} S$
 $B \leftarrow S : u_c$
 $A \rightarrow B : m, i, le$
 $v = 7$
 $lp = 1$
 $t_L = 0x00$
LOOP : B compute
 $A \rightarrow B : p, h, a$
 $r = R(384)$
 $n_S = r[0 : 128]$
 $n_{IV} = r[128 : 128 + n_{IV}l]$
 $n_{IVH} = D_H(2, kp_H, H(n_S, a, v, lp, n_{IV}))$
 $k_H = D_H(1, kp_H, H(n_S, a, v, lp, u_c))$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_c))$
 $k_{M0} = D_P(p \parallel u_c \parallel a \parallel v \parallel lp, n_S, i)$
 $k_C = D_H(1, kp_C, H(n_S, k_{M0}))$
 $m_0 \parallel \dots \parallel m_N = m$
 $cd = cd_0 \parallel \dots \parallel cd_N$
 $m = cd$
 $lp = lp + 1$
 goto *LOOP* if $lp \leq le$
 $A \leftarrow B : cd$

cd_0 Encryption by B

$h_E = E_a(h, n_{IVH}, k_H)$
 $h_El = \text{len}(h_E)$
 $f = 1$ if $\boxed{\text{TERM}}$ else 0
 $ad_F = f \parallel a \parallel n_{IV} \parallel n_S \parallel i \parallel le \parallel lp \parallel h_El \parallel h_E$
 $ad_X = ad_F \parallel k_C$
 $m_E = E_a(m_0, n_{IV}, ad_X, k_{M0})$
 $l = \text{len}(ad_F \parallel m_E)$
 $t = H(v, l, ad_F, m_E, t_L, k_S)$
 $t_L = t$
 $cd_0 = t \parallel v \parallel l \parallel ad_F \parallel m_E$

cd_N Encryption by B

$r = R(384)$
 $n_{IV} = r[0 : n_{IV}l]$
 $f = 1$ if $\boxed{\text{TERM}}$ else 0
 $ad_F = f \parallel a \parallel n_{IV}$
 $ad_X = ad_F \parallel k_C$
 $k_{MN} = D_H(N, kp_B, H(n_S, k_{M0}))$
 $m_E = E_a(m_N, n_{IV}, ad_X, k_{MN})$
 $l = \text{len}(ad_F \parallel m_E)$
 $t = H(v, l, ad_F, m_E, t_L, k_S)$
 $t_L = t$
 $cd_N = t \parallel v \parallel l \parallel ad_F \parallel m_E$

Message Decryption by A

$A \xleftrightarrow{\text{webauthn}} B, M \xleftrightarrow{\text{webauthn}} S$
 $B \leftarrow S : u_c$
 $A \rightarrow B : cd$
 $t_L = 0x00$
LOOP : B compute
 $cd_0 \parallel \dots \parallel cd_N = cd$
 $t, v, l, ad_F, m_E = cd_0$
 $f, a, n_{IV}, n_S, i, le, lp, h_E l, h_E = ad_F$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_c))$
 $b = V(v, l, ad_F, m_E, t_L, k_S, t)$
 $t_L = t$
 if ! b :
 $A \leftarrow B : err$
 $n_{IVH} = D_H(2, kp_H, H(n_S, a, v, lp, n_{IV}))$
 $k_H = D_H(1, kp_H, H(n_S, a, v, lp, u_c))$
 $h = E_a^{-1}(h_E, n_{IVH}, k_H)$
 $B \rightarrow A : h$
 $B \leftarrow A : p$
 $k_{M0} = D_P(p \parallel u_c \parallel a \parallel v \parallel lp, n_S, i)$
 $k_C = D_H(1, kp_C, H(n_S, k_{M0}))$
 $m = m_0 \parallel \dots \parallel m_N$
 $cd = m$
 goto *LOOP* if $lp > 1$
 $A \leftarrow B : m$

m_0 Decryption by B

$ad_X = ad_F \parallel k_C$
 $m_0 = E_a^{-1}(m_E, n_{IV}, ad_X, k_{M0})$

m_N Decryption by B

$t, v, l, ad_F, m_E = cd_N$
 $f, a, n_{IV} = ad_F$
 $b = V(v, l, ad_F, m_E, t_L, k_S, t)$
 $t_L = t$
if ! b :
 $A \leftarrow B : err$
 $ad_X = ad_F \parallel k_C$
 $k_{MN} = D_H(N, kp_B, H(n_S, k_{M0}))$
 $m_N = E_a^{-1}(m_E, n_{IV}, ad_X, k_{MN})$

u_c Encryption at Rest Variables

a - Symmetric AEAD cipher and mode: 2
 ad_F - Additional data included in ciphertext
 ad_X - ad_F with extended context
 b - Valid or invalid MAC tag
 cd - Cipher data
 err - Error message and exit
 f - Block flags: 1 (fixed)
 h_{El} - Encrypted hint length: 0 (unused)
 i - PBKDF2 iteration count: 0 (unused)
 k_C - 256 bit ephemeral commit key
 k_{KS} - HMAC-SHA-512 non-extractable CryptoKey, stored in IndexedDB
 k_L - 256 bit ephemeral per session master key
 k_M - 256 bit ephemeral message cipher key
 k_S - 256 bit ephemeral MAC key
 kp_C - Key purpose text: "Cmit_Key"
 kp_M - Key purpose text: "Cphr_Key"
 kp_S - Key purpose text: "Sign_Key"
 l - Payload length
 le - Loop end: 1 (fixed)
 lp - Loop count: 1 (fixed)
 m_E - Encrypted message
 n_{IV} - Pseudo random initialization vector
 $n_{IV}l$ - n_{IV} bit length: 192
 n_S - 128 bit pseudo random salt
 $pkId$ - Passkey credential identifier
 r - 320 bits of pseudo random data
 s - IndexedDB slot label: "user-cred-key"
 t - 256 bit MAC tag
 u_c - 256 bit user credential (plaintext)
 u_i - User id
 v - Cipher data version: 7

k_L Derivation by B

$B \leftarrow M : pkId$
 $k_{KS} = G()$
 $k_L = D_S(s \parallel pkId, k_{KS})[32 : 64]$

u_c Encryption by B

$B \leftarrow S : u_c, u_i$
 $r = R(320)$
 $n_S = r[0 : 128]$
 $n_{IV} = r[128 : 128 + n_{IV}l]$
 $k_M = D_H(0, kp_M, H(n_S, a, v, lp, u_i, k_L))$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_i, k_L))$
 $k_C = D_H(1, kp_C, H(n_S, k_M))$
 $ad_F = f \parallel a \parallel n_{IV} \parallel n_S \parallel i \parallel le \parallel lp \parallel h_{El}$
 $ad_X = ad_F \parallel u_i \parallel k_C$
 $m_E = E_a(u_c, n_{IV}, ad_X, k_M)$
 $l = len(ad_F \parallel m_E)$
 $t = H(v, l, ad_F, m_E, 0x00, k_S)$
 $cd = t \parallel v \parallel l \parallel ad_F \parallel m_E$

u_c Decryption by B

$B \leftarrow S : u_i$
 $B \leftarrow \text{session storage} : cd$
 $t, v, l, ad_F, m_E = cd$
 $f, a, n_{IV}, n_S, i, le, lp, h_E l = ad_F$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_i, k_L))$
 $b = V(v, l, ad_F, m_E, 0x00, k_S, t)$
if ! b :
 $A \leftarrow B : err$
 $k_M = D_H(0, kp_M, H(n_S, a, v, lp, u_i, k_L))$
 $k_C = D_H(1, kp_C, H(n_S, k_M))$
 $ad_X = ad_F \parallel u_i \parallel k_C$
 $u_c = E_a^{-1}(m_E, n_{IV}, ad_X, k_M)$

Authentication Variables

u_n - A's chosen user name
 u_i - 128 bit user id guaranteed to be unique
 u_r - 128 bit recovery id
 u_c - 256 bit user credential
 o - Quick Crypt origin "https://quickcrypt.org"
 ch - 256 bit challenge value
 ro - Registration options, including o, ch, u_i
 rr - Registration response, including signed ch
 ao - Authentication options, including o, ch
 ar - Authentication response, including signed ch
 cw - Alice's webauthn authenticator credentials

Registration by A

$A \rightarrow B : u_n$
 $B \rightarrow S : u_n, o$
 S create and store :
 $u_i = R(128)$
 $ch = R(256)$
 $B \leftarrow S : ro$
 $B \rightarrow M : ro$
 $A \rightarrow M : cw$
 M create and store passkey, sign ch
 $B \leftarrow M : rr$
 $B \rightarrow S : rr, u_i, ch$
 S verify signature, store rr , generate u_c and u_r , remove ch
 $u_c = R(256)$
 $u_r = R(128)$
 $B \leftarrow S : u_i, u_n, u_r, u_c$
 $A \leftarrow B : u_r \parallel u_i$ as BIP39

Authentication by A

$B \rightarrow S : o[, u_i]$
 S create and store :
 $ch = R(256)$
 $B \leftarrow S : ao$
 $B \rightarrow M : ao$
 $A \rightarrow M : cw$
 M sign ch
 $B \leftarrow M : ar$
 $B \rightarrow S : ar, ch$
 S verify signature, remove ch
 $B \leftarrow S : u_i, u_n, u_c$

Recovery from Lost Passkey by A

$A \rightarrow B : u_r \parallel u_i$ as BIP39

$B \rightarrow S : u_i, u_r, o$

S verify u_r , delete existing rr , create and store :

$ch = R(256)$

$B \leftarrow S : ro$

$B \rightarrow M : ro$

$A \rightarrow M : cw$

M create and store passkey, sign ch

$B \leftarrow M : rr$

$B \rightarrow S : rr, u_i, ch$

S verify signature, store rr , remove ch

$B \leftarrow S : u_i, u_n, u_c$